

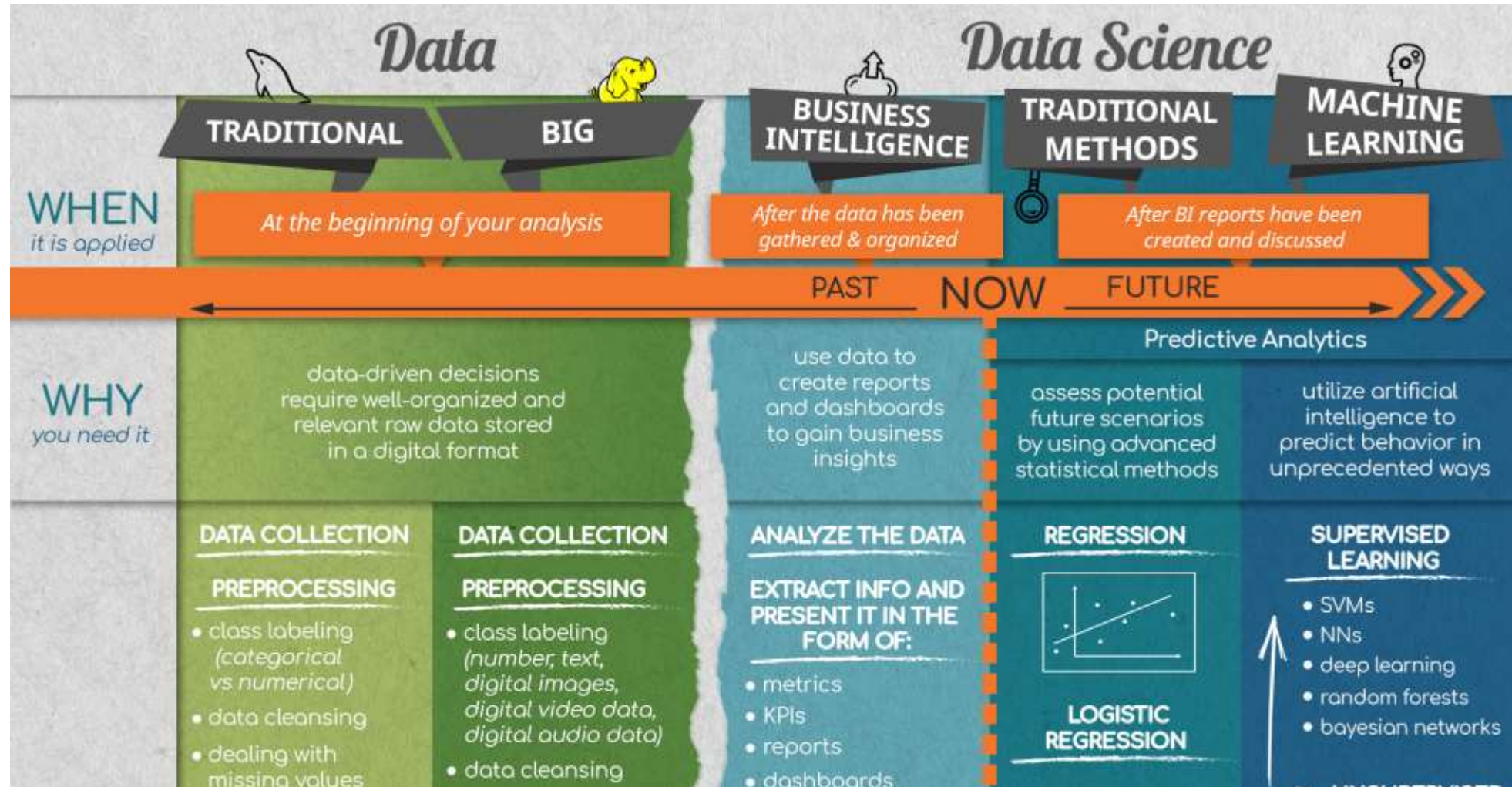
DATA SCIENCE

Techniques for Working with Traditional Data

Data collection, pre-processing, and the case-specific approaches that turn raw facts into information you can actually use.

Where Traditional Data Sits on the Map

Five columns, one timeline. Everything in this deck lives in the very first one.



So What Exactly Is “Data”?

One word, three different meanings. That is why we separate the phrases before talking about techniques.



Data

= raw facts ?

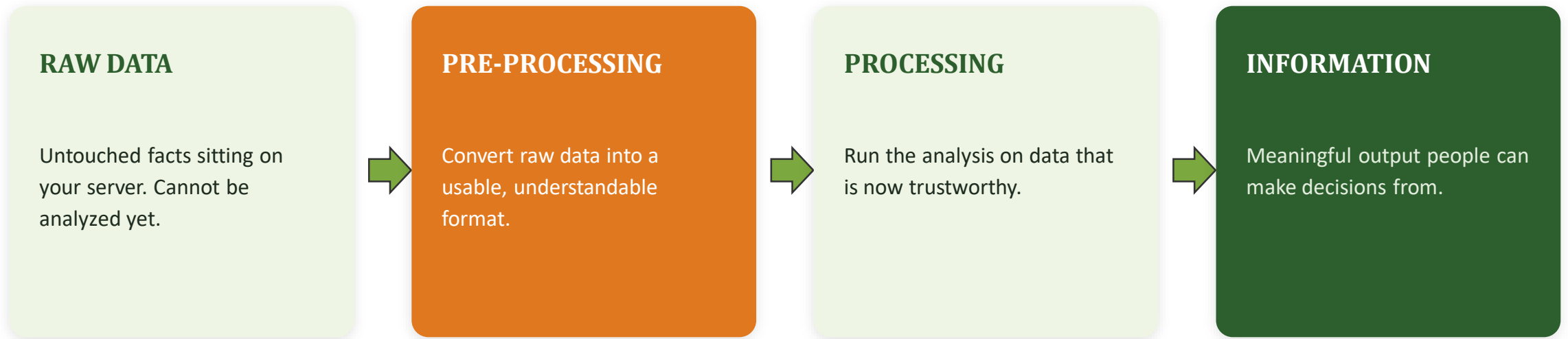
= processed data ?

= information ?

All three answers are correct, and that is exactly the problem. Splitting the term apart is what makes the techniques in this section easy to place.

From Raw Data to Meaningful Information

Now that the word is split apart, here is the pipeline every technique sits inside.



People gather raw data that must be processed to obtain meaningful information. The gathering step is called data collection. The clean-up step in between is called data pre-processing.

Raw Data, in Plain Terms

Three names for the same thing, and three facts that define it.

raw data

=

raw facts

=

primary data

1

Cannot be analysed straight away

Nothing has been checked, corrected, or labeled yet. Running an analysis on it now would just produce confident nonsense.

2

It sits on your server, untouched

Data you have accumulated and stored exactly as it arrived, in whatever shape the source handed it to you.

3

Gathering it is called data collection

That is the very first step of the process, and it happens before a single pre-processing operation runs.

Every technique in this section exists because raw data arrives in this state and cannot be used as-is.

Step 1 — Data Collection

Two everyday ways companies accumulate raw facts.

Raw data (raw facts, primary data) is data you have accumulated and stored but cannot analyze straight away.

MANUAL

Surveys

Ask people directly. For example, rate how much you liked a product or an experience on a scale of 1 to 10.

You control the questions, but people answer selectively.

AUTOMATIC

Cookies

Collected without anyone typing anything. Cookies hand companies detailed information about a user's activity on a website.

Huge volume, but messy and full of surprises.

Both give a good idea of how to improve customer satisfaction — but not yet. The data still has problems in it.

Why Pre-process? Because Raw Data Lies

Real examples of what shows up in freshly gathered customer data.

AGE

932

No customer is 932 years old. Someone mistyped, or a system wrote garbage into the field.

NAME

United Kingdom

That is a country, not a person. The value landed in the wrong column.

Before proceeding with any type of analysis, data like this must be marked as invalid or corrected.

That is what data pre-processing is all about — a group of operations that fixes the problems data gathering inevitably creates.

Technique 1 — Class Labeling

Label each data point with its correct type, or arrange the data by category.

NUMERICAL

Numbers that can be manipulated mathematically.

EXAMPLES

- Number of goods sold daily
- Average goods sold per day
- Average goods sold per month

Can you do math on it? YES

CATEGORICAL

Information that cannot have mathematical manipulations applied to it.

EXAMPLES

- A person's profession
- A person's place of birth
- Product category

Can you do math on it? NO

Technique 2 — Data Cleansing

Also called data cleaning or data scrubbing. The goal: deal with inconsistent data.

1 Inconsistent values

You are handed a dataset of US states and a quarter of the names are misspelled. Techniques must be applied to correct those mistakes before anything else happens.

2 Missing values

Not every customer gives you everything you ask for. A common case: you get a name and an occupation, but no age.

So what do you do with the missing age?

Option A

Disregard the customer's entire record.

Option B

Enter the average age of the remaining customers.

Neither answer is automatic. Both cleansing and missing values must be resolved before the data can be processed further.

Case-Specific Approach — Balancing

When one group dominates the sample, your trends belong to that group, not to everyone.

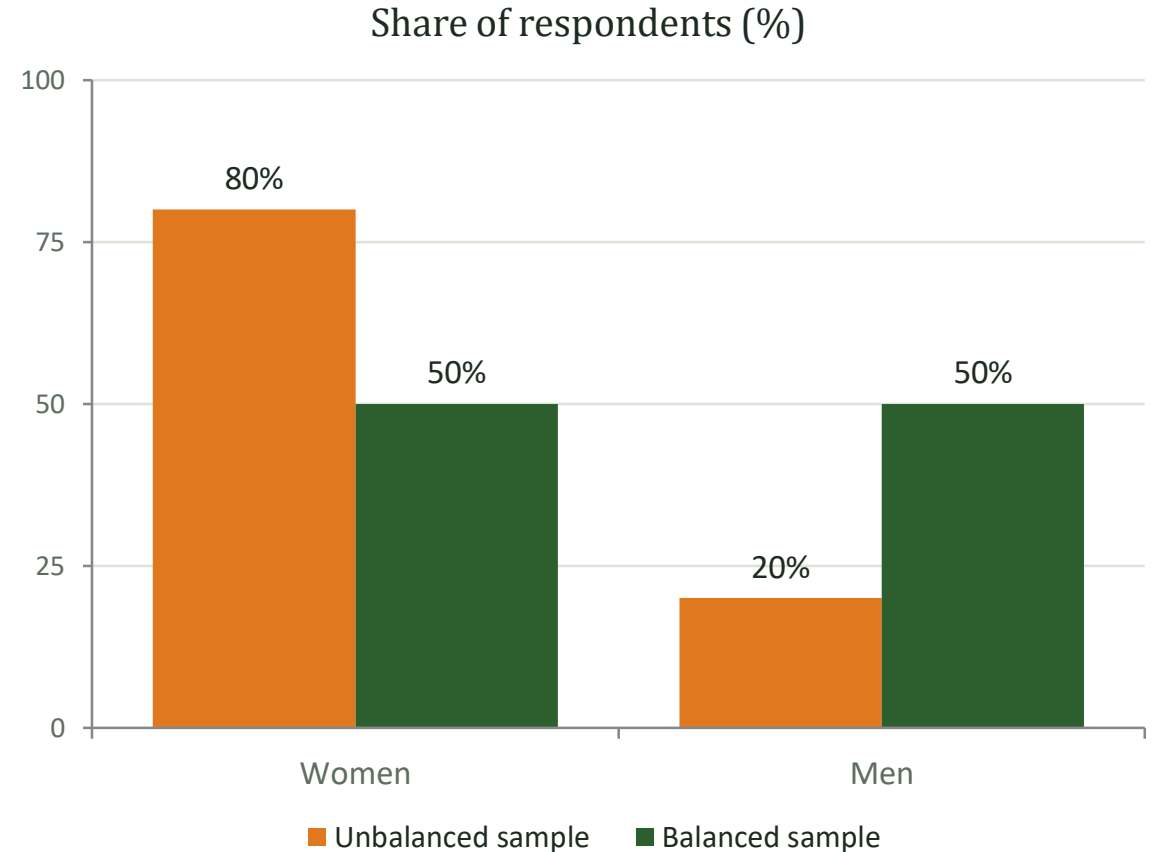
The problem

You run a survey on the shopping habits of men and women, hoping to see who spends more during the weekend.

Then you check who actually answered: 80% female, 20% male. Any trend you spot leans toward women simply because there are four times as many of them.

The fix

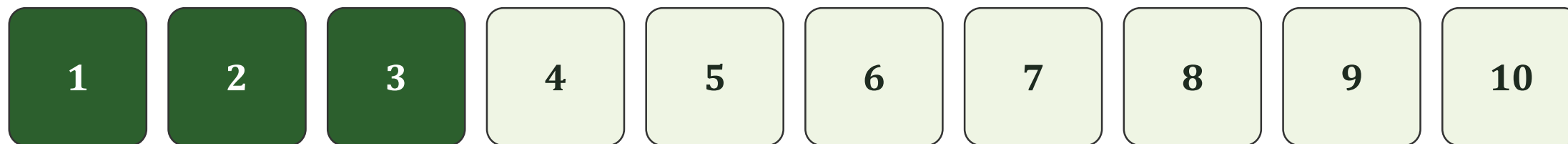
Apply balancing techniques — for example, take an equal number of respondents from each group so the ratio becomes 50-50.



Case-Specific Approach — Data Shuffling

Shuffling your observations is just like shuffling a deck of cards.

COLLECTED ORDER



AFTER SHUFFLING



Take the first 100 rows of an unshuffled dataset and that is not a random sample — whatever was entered first is simply what gets extracted first.

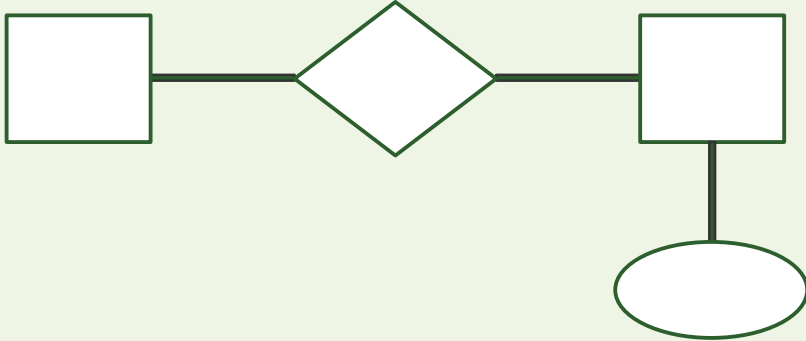
Shuffle first and any 100 consecutive entries are random and most likely representative. It improves predictive performance and avoids misleading results.

How Traditional Databases Are Pictured

Two visualizations you will keep meeting around relational database management systems.

Entity Relationship Diagram

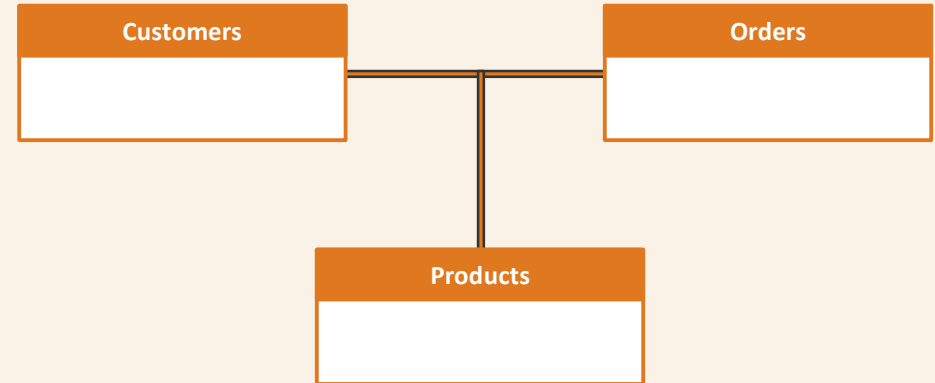
A complex, theoretical way of illustrating a database architecture. Specialists like it because the shapes convey directly how tables relate.



entity — relationship — entity, with attributes attached

Relational Schema

Important to understand well — not only by database creators, but by the people who use the database every day.



each rectangle is one data table, each line shows which tables relate

Traditional Data at a Glance

The full column from the infographic, in one place.

WHEN IT IS APPLIED

At the very beginning of your analysis, before anything is reported or predicted.

WHY YOU NEED IT

Data-driven decisions require well-organized, relevant raw data stored in a digital format.

WHAT TECHNIQUES

Data collection, pre-processing (class labeling, cleansing, missing values), plus balancing and shuffling.

WHERE IT SHOWS UP

Basic customer data. Historical stock price data.

HOW — THE TOOLS

R, Python, SQL, MATLAB. Excel and IBM SPSS.

WHO DOES THE WORK

Data architect. Data engineer. Database administrator.

Are you aware? 200,000 lines of data is not necessarily big data. Volume alone does not make a dataset “big” — variety, variability, velocity and veracity matter too.

Example 1 — Basic Customer Data

The same table type used in a SQL course, and a clean way to see the two labels in action.

Customer ID	Name	Country	Complaints filed
1001	Kim Bradley	United Kingdom	3
1002	Jose Reyes	Philippines	0
1003	Anna Kruger	Germany	5
1004	Liam Chen	Singapore	1

CATEGORICAL

Customer ID

They look like numbers, but an average customer ID tells you nothing. They carry no numerical value, so they are categorical.

NUMERICAL

Complaints filed

Add them together and you get the total number of complaints — useful information. These are genuinely numerical.

The test is never how the value looks. It is whether the arithmetic means anything.

Example 2 — Daily Historical Stock Prices

The dataset used in a Python for Finance course. Same lesson, completely different domain.

Date	Ticker	Closing price (USD)
2024-01-03	AAPL	184.25
2024-01-04	AAPL	181.91
2024-01-05	AAPL	181.18
2024-01-08	AAPL	185.56
2024-01-09	AAPL	185.14
2024-01-10	AAPL	186.19

CATEGORICAL

Date

The dates of the observations. They order the rows, but you would never average them.

NUMERICAL

Closing price

Averages, returns, volatility — every one of those calculations is meaningful here.

One column of dates, one column of prices — and the exact same categorical-versus-numerical split you saw in the customer table.

Why Both Count as Traditional Data

Two unrelated domains, one shared structure. That structure is what makes them “traditional.”

1

Not structurally complex

Plain rows and columns. Nothing nested, no images, no audio, no free-form text to untangle first.

2

Small enough for tables

Neither dataset is too large to organize into an ordinary table on an ordinary machine.

3

Clear and understandable

Every field means something obvious on sight. No decoding step before the work can start.

Where you have already seen these

Basic customer data and historical stock price data are exactly the two entries listed under WHERE in the Traditional column of the infographic. Everything covered in this section applies to both.

PART 2

Big Data

Same steps carry over. The variety explodes, and everything gets harder to make sense of.

Same Backbone, More Variety

Big data does not throw the pipeline away. It stretches every step of it.

What carries over

- Data collection
- Pre-processing before any analysis
- Grouping data into classes or categories
- Data cleansing
- Dealing with missing values

What changes

- Far more data types than numerical and categorical
- A much wider range of cleansing methods
- Missing values become a far bigger problem
- New case-specific techniques: text data mining, data masking

Collecting and pre-processing big data is essential to organize it before running any analysis or making any prediction.

Class Labeling Gets a Lot Wider

The step is the same. There are simply far more classes to label into.

Numerical

the traditional pair

Categorical

the traditional pair

Text data

articles, papers, posts

Digital image data

photos, scans, frames

Digital video data

recordings, streams

Digital audio data

calls, clips, speech

Green: what traditional data already gave you. Orange: what big data adds on top.

More Types Mean More Ways to Clean

A wider range of data types drags a wider range of cleansing methods along with it.

Cleansing methods multiply



Techniques that verify a digital image observation is actually ready for processing.



Approaches that confirm the audio quality of a file is adequate before you proceed.

Missing values

Big data has big missing values, which is a big problem.

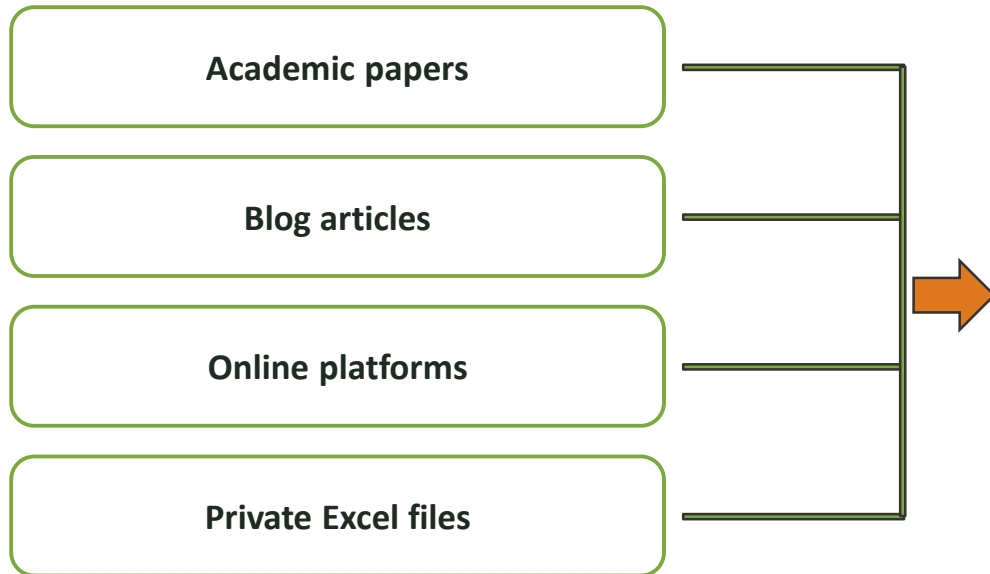
The step you met on traditional data does not get easier at scale. It gets louder.

Same three pre-processing jobs as before: label the classes, cleanse the data, and decide what to do about whatever is missing.

Case-Specific — Text Data Mining

The process of deriving valuable unstructured data from text.

The research question: pull every mention of marketing expenditure out of everything you have stored.



Text data mining

With few sources and a small volume of text, you could find what you need by hand without much of a problem.

When the volume is huge and spread across every format above, that stops being possible. Which is why academics and practitioners built methods to do the extraction for you.

Case-Specific — Data Masking

Analyze the information without ever exposing the private details inside it.

Why it exists

Credible business and governmental activity depends on preserving confidential information. Personal data being shared online does not make it fair game to expose.

How it works

It conceals the original data with random and false data, so the analysis can still run while everything confidential stays locked away. Like shuffling, it can get quite complex.

What the analyst actually sees

ORIGINAL RECORD

Name: Anna Kruger
Email: a.kruger@example.com
Account: 4519 8823 0067

MASKED RECORD

Name: Q7RTX Vbnm
Email: x8k2@masked.local
Account: **** * 0067

What a Big Data Visualization Looks Like

A simplified version of what big data teams deal with constantly.

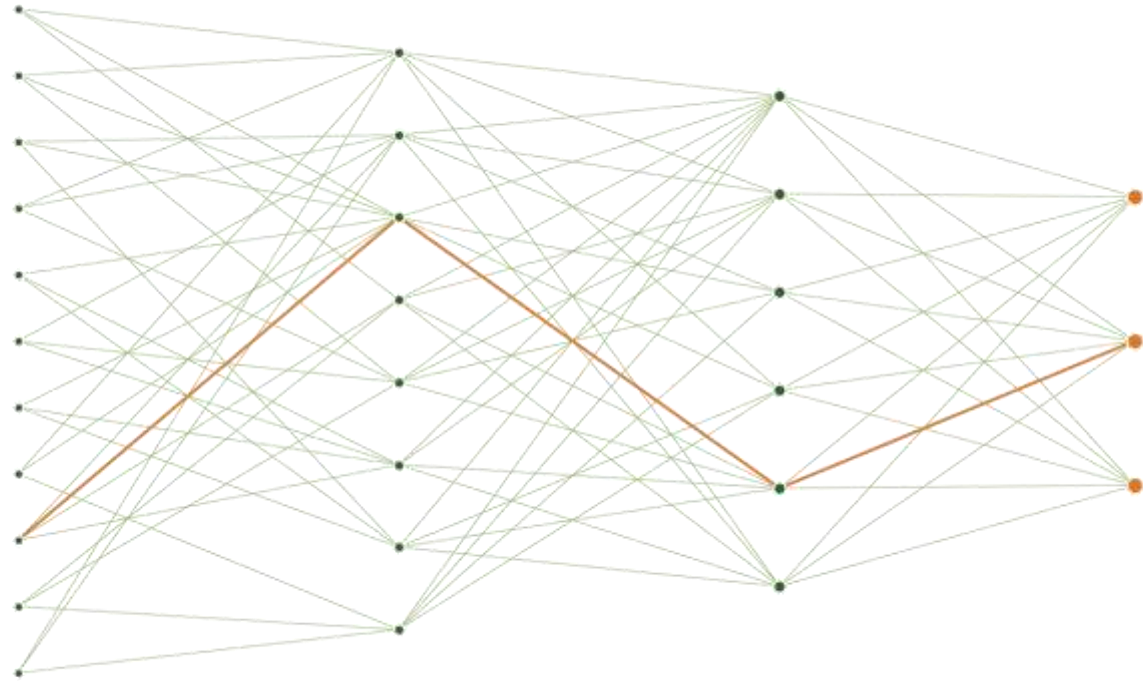
Traditional data

The objects on the diagram represent data tables and the relationships between them. You can trace it by eye.

Big data

The objects show the complex way data is derived from many different sources. Tracing any single value means following it back through layers.

Rule of thumb: a large number of intersecting lines almost always means you are looking at big data.



Many sources on the left, a handful of derived values on the right, and very little you can follow by eye.

Where Big Data Shows Up — Social Platforms

One of the largest online communities on earth, and a textbook case of all three Vs at once.

What gets stored: user names, personal data, photos, videos, recorded messages, and more besides.

VARIETY

Every data type at once

Text, images, video, audio and structured profile fields all sitting in the same place. Numerical and categorical does not begin to cover it.

VOLUME

2 billion+ users

With that many people worldwide, the sheer amount of data held on their servers is tremendous by any measure.

VELOCITY

Reported in real time

Real-time reporting of the aggregated, anonymized voice of its users, plus analytical tools running across its mobile applications.

That last V is why the company keeps investing in real-time data processing power. It is buying velocity.

Where Big Data Shows Up — Financial Trading

The same stock prices you met earlier. Change the sampling rate and they stop being traditional data.

What happens if you record the price every five seconds? Or every single second?

DAILY CLOSE

1

observation per stock, per day

traditional data

EVERY 5 SECONDS

4,680

observations per stock, per day

big data

EVERY SECOND

23,400

observations per stock, per day

big data

Based on one 6.5-hour trading session, for a single stock.

A dataset like this is incredibly voluminous.

It demands significantly more memory and disc space, and a different set of techniques to pull any meaningful information back out of it.

PART 3

Business Intelligence

The data is organized. Now explain what actually happened, and why anyone should care.

You Have Left Raw Data Behind

Pre-processing is done. The data is organized and every entry means something you can read.

Business intelligence is where your data skills combine with your business knowledge and intuition to explain your company's past performance with conviction.

Questions you can now answer without flinching

What happened?

When did it happen?

**How many units did
we sell?**

**In which region did
we sell the most
goods?**

Every one of these is a question about the past. That is exactly the territory business intelligence covers.

One Problem, Every Angle

A good BI analyst never looks at a problem from only one direction.

THE SIMPLE ASK

“What's the news regarding our email marketing?”

Quick, open-ended, and answered in a sentence if you know where the numbers live.

THE COMPLEX ASK

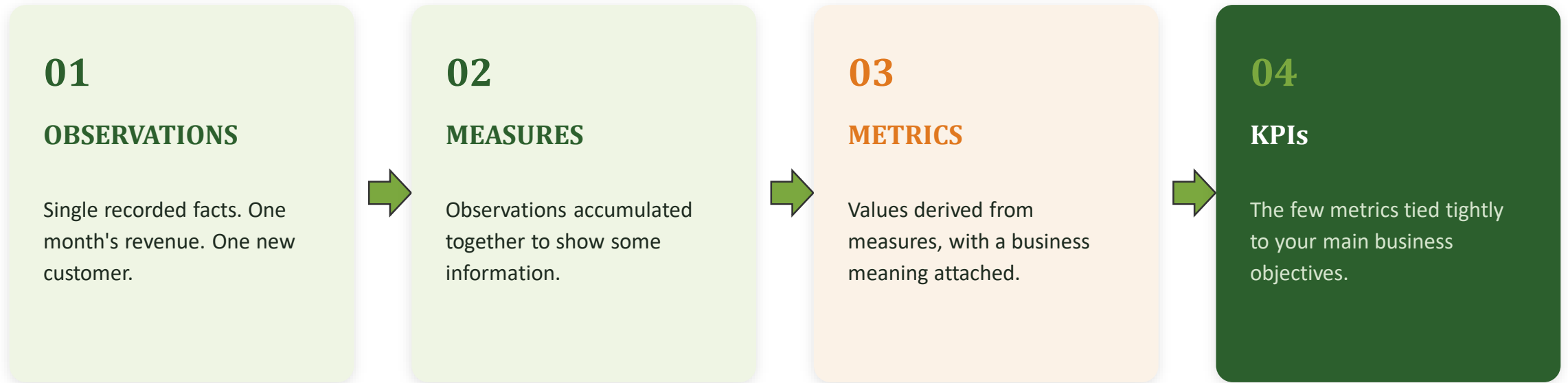
“How did our email marketing perform last quarter in terms of click-through rates and revenue generated, and how does that compare to the same quarter last year?”

Multiple measures, a defined timeframe, and a year-over-year comparison, all in one breath.

The job is to understand the essence of a business and then strengthen that business through the power of data.

Observations → Measures → Metrics → KPIs

Four words that get used interchangeably in conversation and mean four different things at work.



Observations lead to measures. Measures create metrics. Metrics are the ones with business applications attached.

The Chain, With Real Numbers

Quantification is the step that turns observations into something you can do arithmetic on.

Quantification: representing observations as numbers. Without it, no mathematical manipulation is possible at all.

Month	Revenue	New customers
January	\$100	10
February	\$120	15
March	\$130	25
Q1 measure	\$350	50

THE METRIC

$$\$350 \div 50 = \$7$$

average quarterly revenue per new customer

Not just a number. A number carrying a business meaning.

A measure is the accumulation of observations. Total the three months and you have one.

Metrics exist for comparison. Track this one every three months and you can compare customer retention and spending quarter by quarter.

From Metric to KPI

A real business yields hundreds of metrics. Tracking them all is possible. It is just not sensible.

KEY

Related to your main business goals, not to whatever happened to be measurable.

PERFORMANCE

Shows how successfully you performed within a specified timeframe.

INDICATORS

Values that point at something real about your business performance.

A METRIC

Traffic to a page on your website, counted from any type of user who happened to land there.

A KPI

Traffic to that same page, counted only from users who clicked the link in your ad campaign.

Only the second one tells you whether the ads are actually motivating clicks, and therefore whether to keep paying for them.

Every Number You Extract Must Be Seen

Calculating metrics is not the whole job. Making them legible to other people is.

General managers want dashboards and reports: easily digestible visuals, supported by the most relevant numbers.

Graphs

Diagrams

Maps

Dashboards

The filtering is the skill

Dropping the boring metrics and turning the interesting, informative KPIs into comparable visuals is a core part of the analyst's job.

Why KPI dashboards exist

They look much like metric dashboards. The difference is that they keep executives and managers focused on the main business objectives.

On the infographic, this is the whole Business Intelligence column: analyze the data, then present it as metrics, KPIs, reports and dashboards.

Where BI Shows Up — Price Optimization

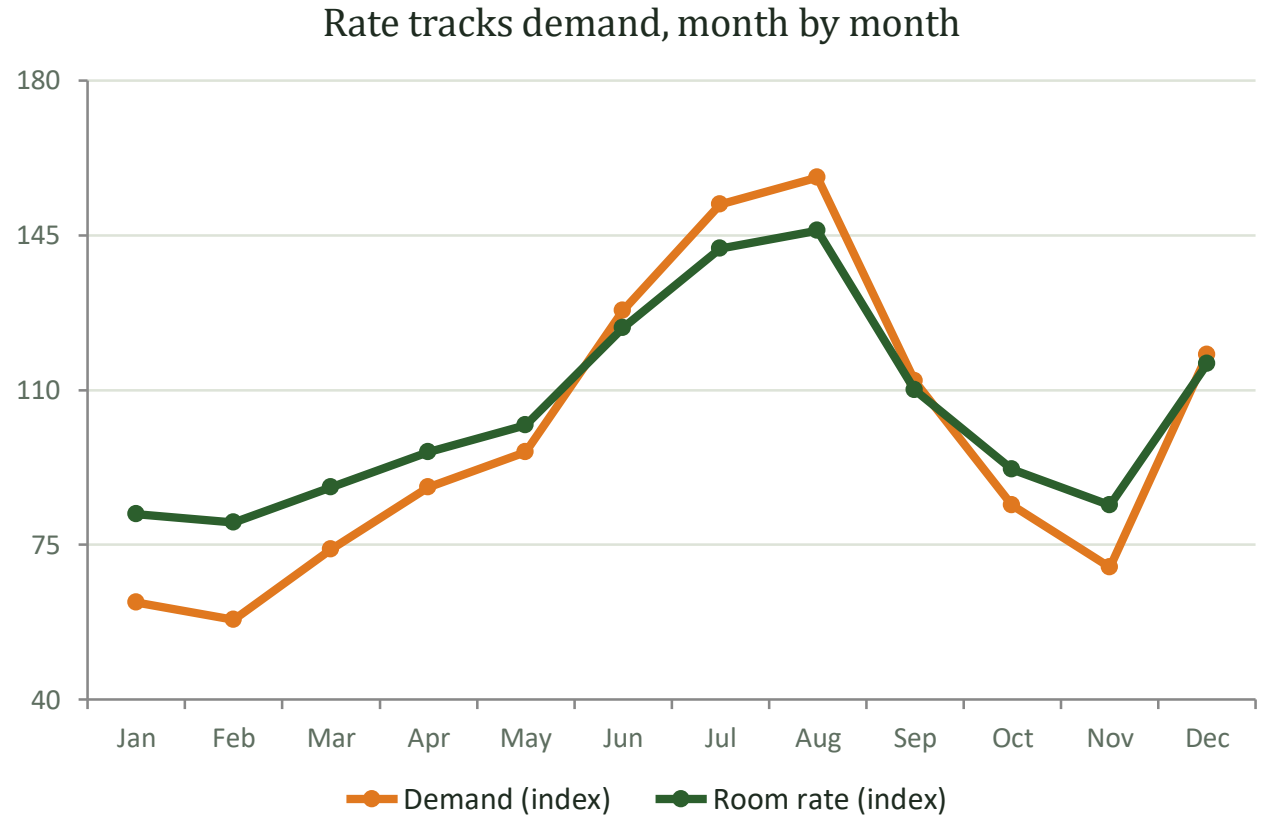
Hotels run this play constantly, and it works because the pricing follows the data.

The play

Raise the room rate in the periods when plenty of people want to visit. Drop it to attract visitors when demand falls away. Done well, profit rises sharply.

The requirement

You have to pull the relevant information in real time and compare it against historicals. BI is what lets you adjust the strategy to past data the moment that data exists.



Illustrative figures, both indexed to make the relationship visible on one axis.

Where BI Shows Up — Inventory Management

Both directions hurt. The whole game is landing between them.

OVERSUPPLY

Stock nobody buys. Waste, storage cost, and cash tied up in a warehouse.

UNDERSUPPLY

Demand you cannot serve. Lost sales, and customers who go elsewhere next time.

The target: supply enough stock to meet demand, with the minimal amount of waste and cost.

Hit it and shipment logistics run efficiently, which is how cost comes down and profit goes up.

1

Seasonality patterns

An in-depth read of past sales transactions to find the times of year that sell hardest.

2

Over- and under-stocked months

Track inventory back through the year and the mismatched months name themselves.

3

Peak day, even peak hour

A detailed enough analysis pinpoints the day, or the time of day, when the need for a good is highest.

PART 4

Traditional Methods

Classical statistical forecasting. The point where you stop explaining the past and start predicting the future.

From Analysis to Analytics

The reports are built, the executives have their insights. So what do you do with the information?

You use it to predict future values as accurately as you can. That is the moment you stop dealing with analyses and start applying analytics.

Predictive analytics splits into two branches

Traditional methods

Classical statistical methods for forecasting. Regression, logistic regression, cluster analysis, factor analysis, time series.

covered in this section

Machine learning

Supervised, unsupervised and reinforcement learning, all leaning on artificial intelligence rather than classical statistics.

the next section

Regression

A model used for quantifying causal relationships among the variables in your analysis.

How the picture is built

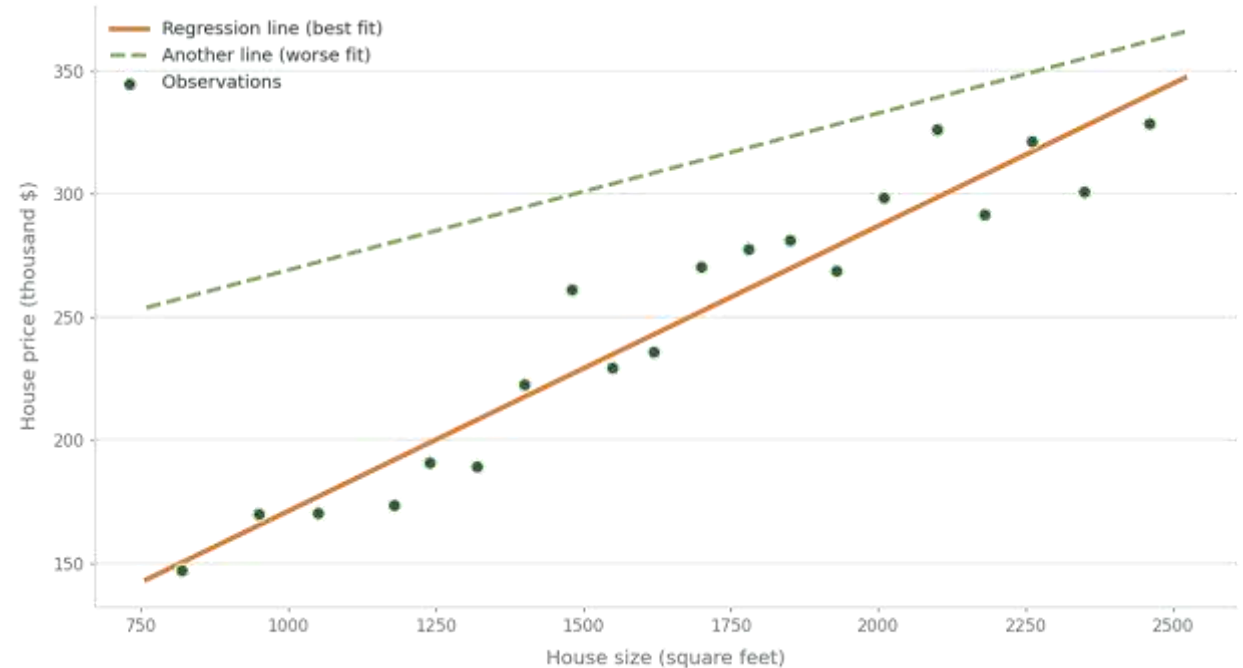
Take a dataset with house prices in one column and house sizes in the other. Every row is one observation, and every observation becomes a dot.

Size runs along the horizontal line, price up the vertical. Further right means a bigger house. Further up means a higher price.

The regression line is the straight line that sits as close as it can to all of the dots at once.

$$Y = B \times X$$

Y is the house price, X is the house size, and B is the coefficient you multiply by. The graph is the visual support for it.



Which line are the observations closer to? The orange one — which is exactly why it represents the distribution more accurately.

Logistic Regression

If linear regression models exist, non-linear ones must too. This is the common example.

What makes it different

Unlike the house prices example, the values on the vertical line are not arbitrary integers. They are ones and zeros, and nothing else.

Where it earns its keep

Decision-making. Companies run logistic regression to filter job candidates during screening.

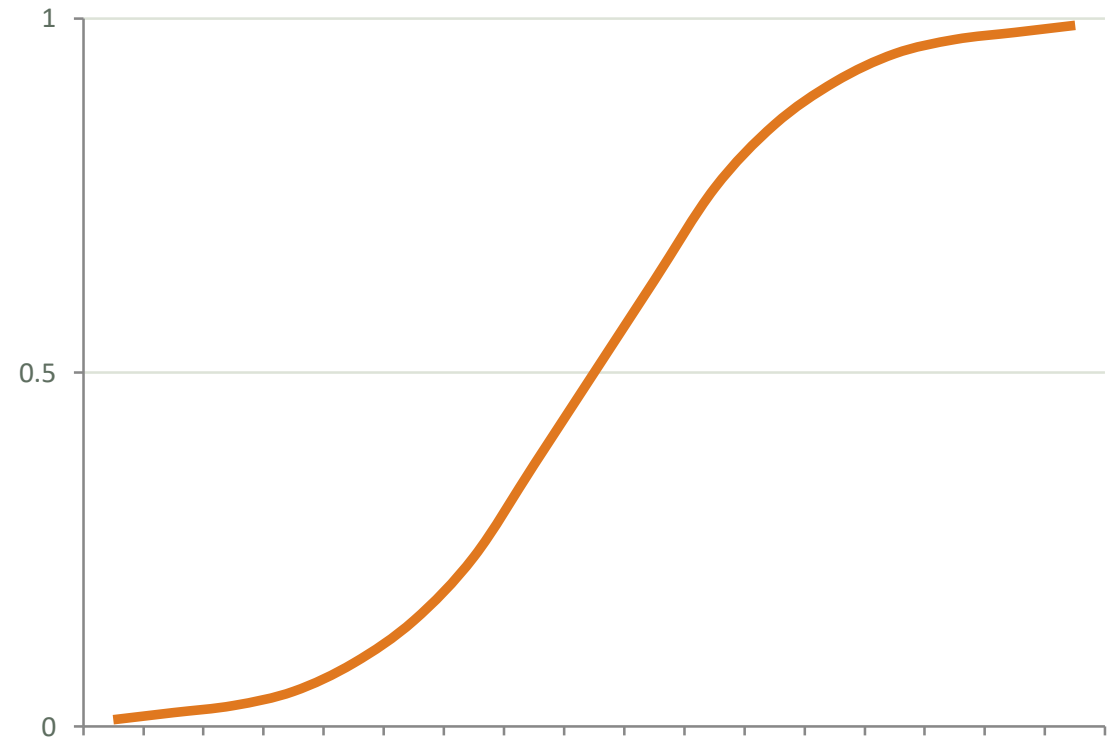
1

Estimated probability of performing well is above 50%

0

Anything at or below that threshold

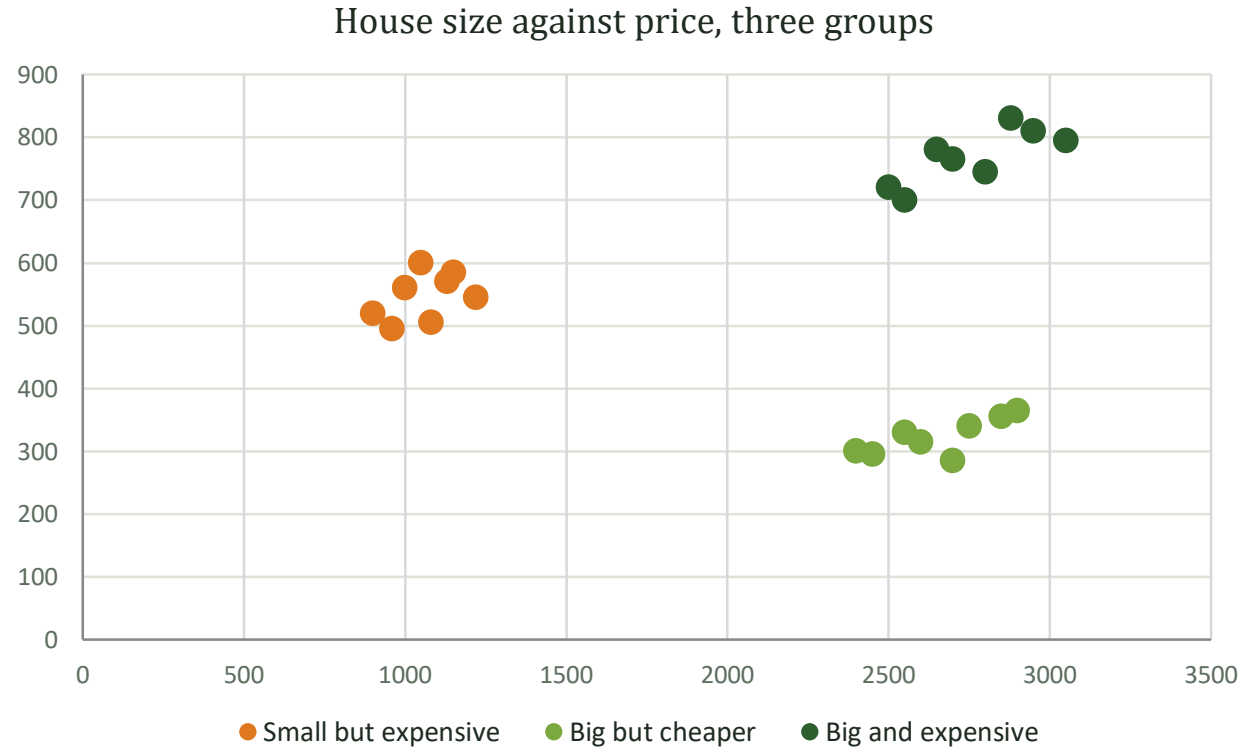
Ones and zeros, joined by an S-curve



Candidate strength runs left to right. The outcome can only ever land on 0 or 1.

Cluster Analysis

A third dataset, dispersed differently. You could draw one regression line — but you can do better.



Small, but expensive
Typical of houses in the city centre.

Big, but cheaper
Far from the city, so the space costs less.

Big and expensive
Probably not central, but still a nice neighbourhood.

Spotting the clusters is what let us conclude that location is a significant factor in pricing a house.

One X Becomes Many

House size is not the only thing that moves a house price, and the maths knows it.

X1

House size

X2

Location

X3

Number of rooms

X4

Year of construction

Once you consider several explanatory variables, the regression model stops having one X and starts having X1, X2, X3, and onward.

Four names for exactly the same thing

Explanatory variable

Regressor

Independent variable

Predictor variable

If a textbook swaps between these four terms mid-chapter, nothing has changed. They all mean the X side of the equation.

Factor Analysis

A survey of 100 questions gives you X1 through X100. Analysing 100 variables is genuinely tough.

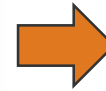
Three questions from that survey

On a scale from 1 to 5, how much do you agree?

1 I like animals.

2 I care about animals.

3 I am against animal cruelty.
Whoever marks 5 on the first will mark 5 on the other two. Nobody strongly agrees with one and disagrees with the rest.



One factor instead of three

General attitude towards animals

Three variables collapse into one without losing what the survey was actually measuring. Do that across the whole thing and 100 variables become roughly 10, which is a size a regression can work with and predict from more accurately.

The one-line distinction worth memorising

Clustering groups observations together. Factor analysis groups explanatory variables together.

Time Series

The technique you will lean on constantly if you work in economics or finance.

What it is

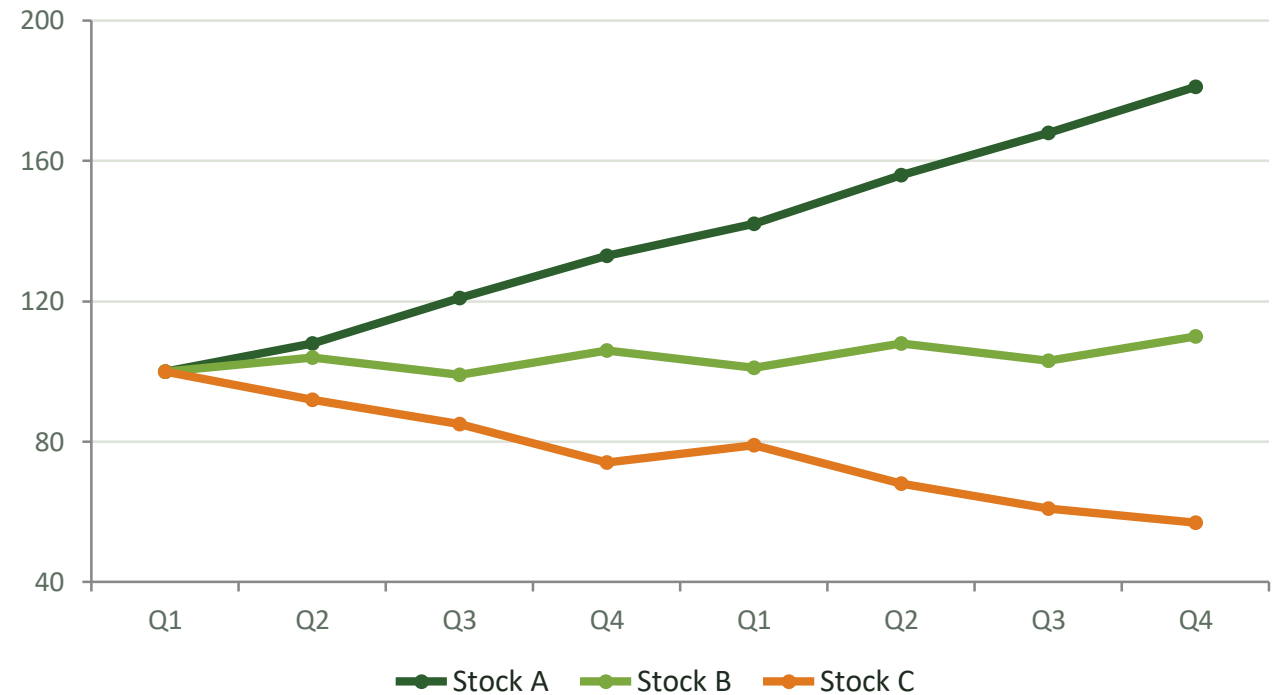
Following the development of certain values over time — stock prices, sales volume, anything that moves.

The one fixed rule

Time always goes on the horizontal line, because time is independent of every other variable in your analysis.

There is a vast variety of methods professionals choose from. This was the brief introduction to the basic types.

Three stocks, two years, one glance



Study the visualisation and you can say immediately which stock performed and which did not.

Where It Shows Up — User Experience

You run UX for a website selling goods globally. Your job is to maximise user satisfaction.

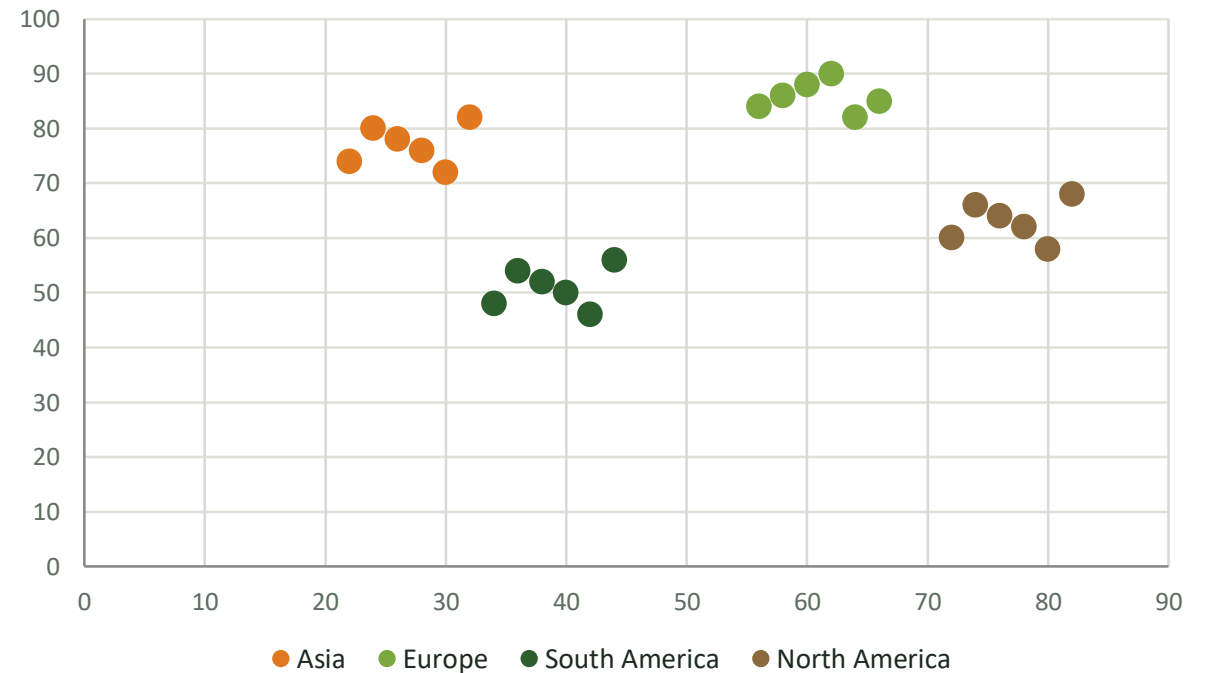
The setup

You designed and ran a survey measuring customer attitude toward the product you just launched globally. Plot the observations and they land like this.

Cluster analysis finds four groups — and each one is a continent



Survey responses: ease of use against satisfaction



Four distinct groups means four separate tests. The gap between clusters is far too wide for one general conclusion.

Where It Shows Up — A/B Testing

Also called hypothesis testing, experimentation, or multivariate testing.

An online company wants to update its website design and has developed two good versions. Which one ships?

GROUP A

Version 1

1,240

signups

GROUP B

Version 2

1,517

signups

ships

Equal-sized groups, one variable changed, and whichever generates more signups wins. With more than two options it becomes A/B/n, or multivariate testing.

Website design

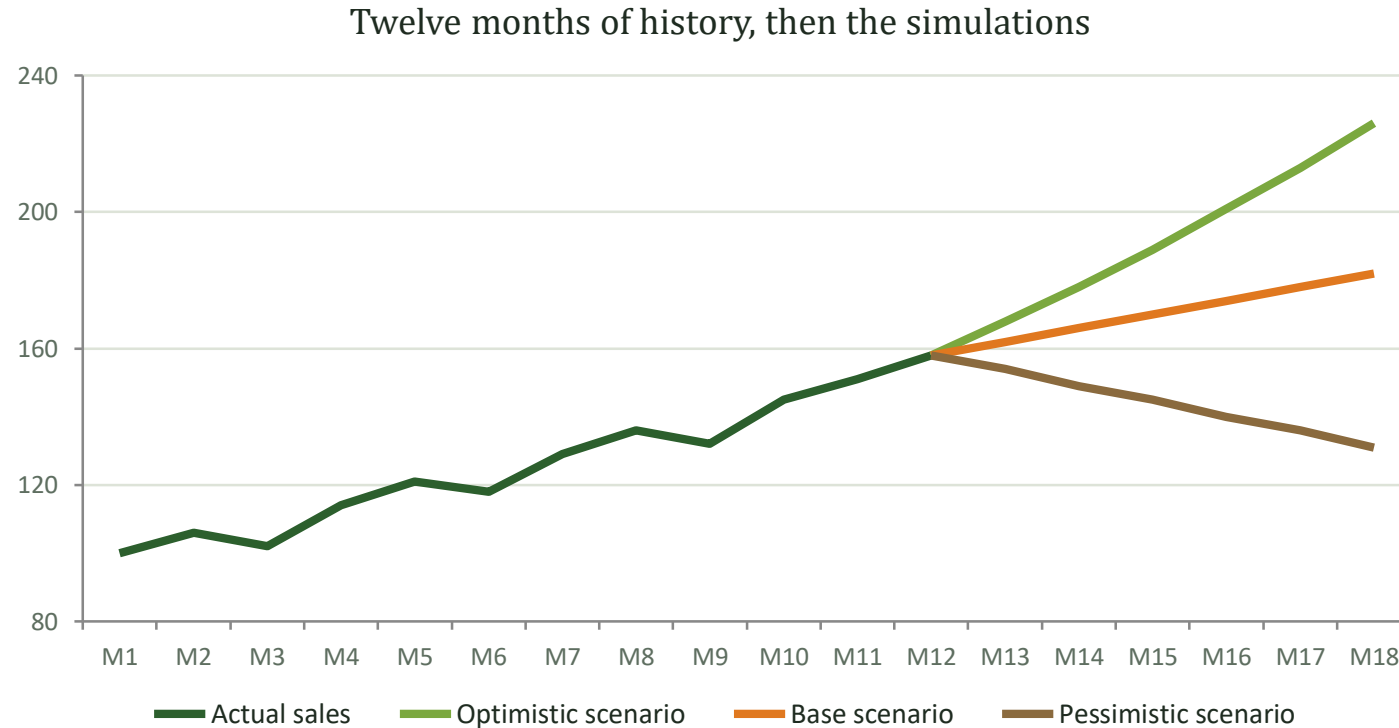
Email marketing

New treatments in medicine

New government policy

Where It Shows Up — Sales Forecasting

Every business and financial company forecasts sales volume. This is the technique underneath it.



The question

This was your data up to a certain date. What happens next? Will the volume rise or fall over the year ahead?

The answer

Several mathematical and statistical models let you run multiple simulations, and each one hands you a future scenario to plan against.

Sales forecasting sits on the frontier with machine learning. ML sharpens and automates the accuracy, but time series analysis is what holds it up.

PART 5

Machine Learning

The other branch of predictive analytics, and the last column on the infographic.

What Sits at the Core of It

Create an algorithm. The computer uses it to find a model that fits the data as well as possible.

Conventional methods

You work out how to find the model, then instruct the machine to follow that route. The thinking is yours. The machine executes it.

Machine learning

You never tell it how to find the model. You give it algorithms that tell it how to learn on its own, then let it get on with it.

So what is a machine learning algorithm, in a few words?

A trial and error process — with one special property. Each consecutive trial is at least as good as the one before it.

The Four Ingredients

Every machine learning setup has these four. The robot archer makes them concrete.

01

Data

The values the model trains on.

The quiver of arrows. All arrows, but straight ones, crooked ones, light ones, heavy ones.

02

Model

The thing being learned and refined.

The usage of the bow. How the robot actually fires it.

03

Objective function

The measure of how wrong you currently are.

How far from the target the shots landed, on average.

04

Optimization algorithm

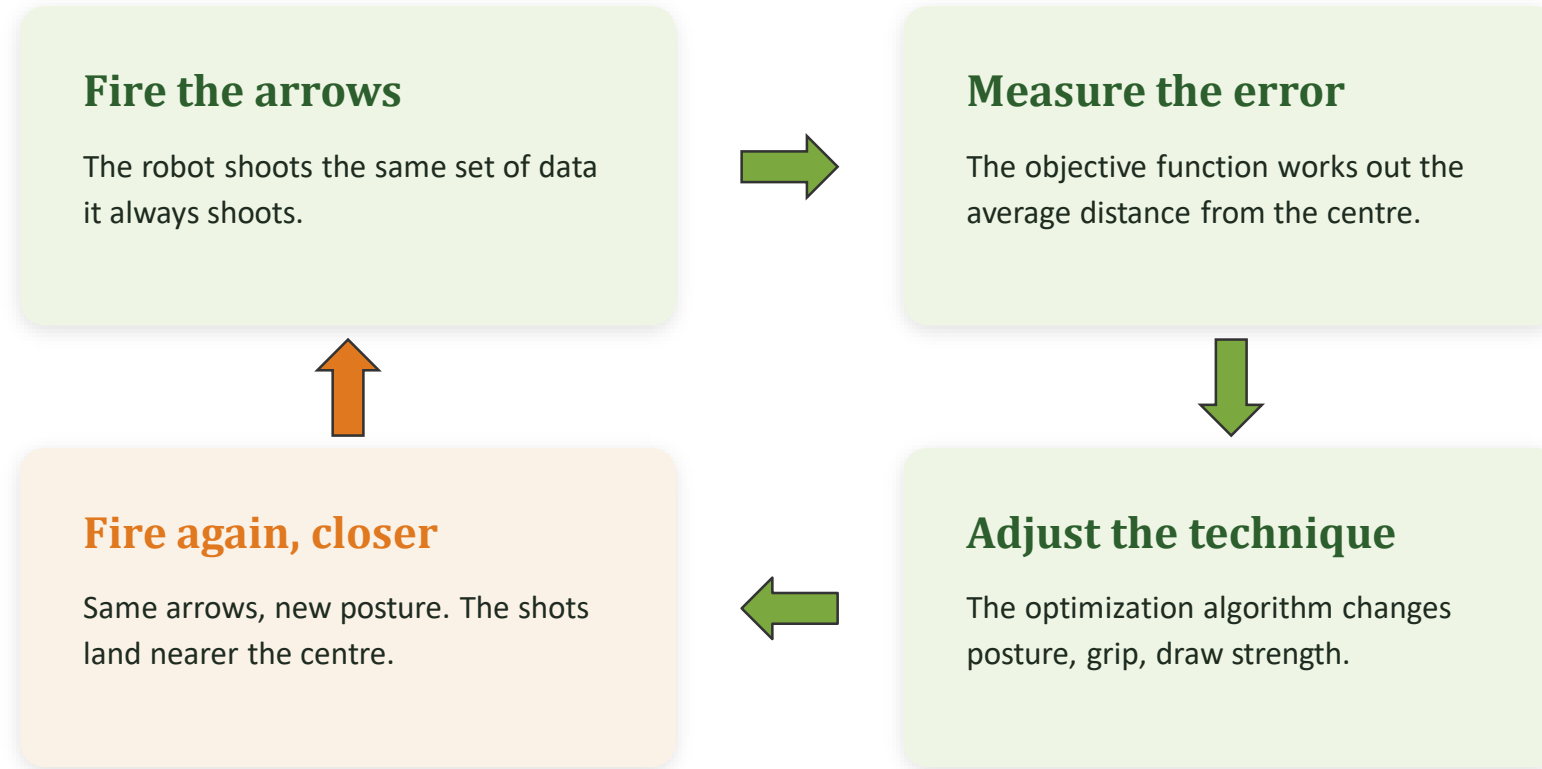
The mechanics that reduce that error.

Adjusting posture, grip, and how hard the bowstring is pulled.

And the target? In supervised learning — the most common type — we already know what we are aiming for. That known thing is the target.

The Loop That Does the Learning

The same arrows, fired again and again, with everything adjusted a fraction each time.



Each pass improves things almost unnoticeably. Run it hundreds or thousands of times and the robot finds the optimal way to fire this set of arrows.

You Give It a Goal, Not a Rulebook

This is the part people find hardest to accept about machine learning.

What you do NOT do

Programme a set of instructions:

- *Place the arrow in the middle of the bow.*
- *Pull the bowstring back.*
- *Line up your elbow, and so on.*

What you DO

Hand it one final goal:

“Place the arrow in the centre of the target.”

Whether it holds the arrow in the middle of the bow or at the bottom is not your problem, as long as it hits.

It will not shoot well straight away. After 100,000 tries, it may be the best archer out there.

So When Does the Training Stop?

There are infinite things it could try. Two reasons it does not try all of them.

1

It carries what it learns forward

The robot picks things up along the way and folds them into the next shots. Once it learns it has to face the target, it stops firing in the opposite direction.

That is the whole purpose of the optimization algorithm.

2

Good enough is genuinely good enough

It cannot fire arrows forever, and hitting the centre nine times out of ten may be all you needed.

Stop it at a chosen accuracy level, or after a chosen number of arrows.

At the end, most shots hit the centre, so the objective function is low — minimised, as we like to say — and the optimization algorithm has done everything it can.

Now Hand It a Different Bag of Arrows

You own a robot that is an amazing archer. The interesting question is what happens next.

IT COPEs

Arrows like the ones it trained on

If it saw most types of arrow during training, it will do great with a fresh bag of them. Expect the centre, or very close to it.

IT STRUGGLES

Half an arrow, or a longer one

Hand it something it has never seen and it simply will not know what to do. Nothing in its training covers that case.

The payoff

The robot can learn to fire more effectively than a human can. It might even work out that we have been holding bows the wrong way for centuries.

Machine Learning Is Not About Robots

The archer was a story. What people actually use machine learning for is improving complex computational models with endless applications in daily life, and especially in business.

And those models are not built from nothing

Regression

The same causal-relationship modelling you met two sections ago.

Cluster analysis

The same grouping of similar observations, at a very different scale.

Taken further

Both pushed to unconventional, and extremely useful, levels.

Still ahead: the three major types of machine learning, and two real-world applications of it.

The Three Major Types of Machine Learning

Same robot, same bow. What changes is what you hand it and what you reward.

Supervised

You know the target for every arrow before it is fired. The most common type by far, and the one the robot archer was doing all along.

labelled data

Unsupervised

No targets at all. You hand over arrows with unknown properties and ask the machine to divide them for you.

unlabelled data

Reinforcement

No error to shrink. A reward to chase instead, handed out whenever the machine does better than last time.

reward system

Supervised learning gets its name honestly: training an algorithm looks a lot like a teacher supervising her students.

Supervised Learning — Labelled Data

There is no single target. Different arrows have targets of their own.

Labelled data means: you know the target before the shot, so you can associate that shot with a target and score it.

Type of arrow	Target it is expected to hit
Small arrow	The closest target
Medium arrow	The target further away
Large arrow	The target hanging on the tree
Crooked arrow	The ground. No target at all

What the robot actually sees

- The ground
- A target close by
- A target further off
- A target hanging on a tree behind
- A house to the side, and the sky

What you supervise is the training itself. If a shot lands far from its target you correct the posture. If it does not, you leave it alone.

Unsupervised Learning — Unlabelled Data

Sometimes you have neither the time nor the resources to give every arrow a target.

What changes

You hand over a bag of arrows with unknown physical properties. Neither you nor the robot has sorted them into groups.

Then you ask it to fire in a direction, with no targets provided. It sees only the ground, the tree, the house and the sky.

So you are no longer after a model that shoots better. You are after one that divides the arrows.

What turns up after thousands of shots

The broken arrows

All fallen on the ground nearby, in one obvious heap.

Small, medium and large

The rest sort themselves out by how far they travelled.

The anomalies

Crossbow bolts in the bag pile up somewhere all of their own.

It fires so many times that it can surprise you — splitting the arrows into five size categories instead of four because it found the crossbow bolts, or flagging which arrows are about to break.

In Practice, You Use Both

Supervised learning could handle those problems too, and often does. It just is not always affordable.

If you have one million arrows, you do not have the time to assign a target to every one of them. That is the whole argument for unsupervised learning.

1 Let it cluster

Give the robot a bag of unclear arrow types and just let it shoot until the data splits into five clusters.

unsupervised

2 Learn the targets

Now that you know which types you have, fire a few shots yourself with each one and work out what it can hit.

you, manually

3 Train it properly

Hand the clustered dataset back with targets attached and let it shoot until the model is precise.

supervised

The two answer different questions, so using them hand in hand gets you further than either on its own.

Reinforcement Learning — The Reward System

Every time the robot fires better than before, it gets a chocolate. Fire worse and it gets nothing.

Instead of minimising an error, you maximise a reward. In other words, you maximise the objective function.

Put yourself in the machine's shoes

REWARD ARRIVES

“I fired an arrow and got a chocolate. What did I do right, so I get more next time?”

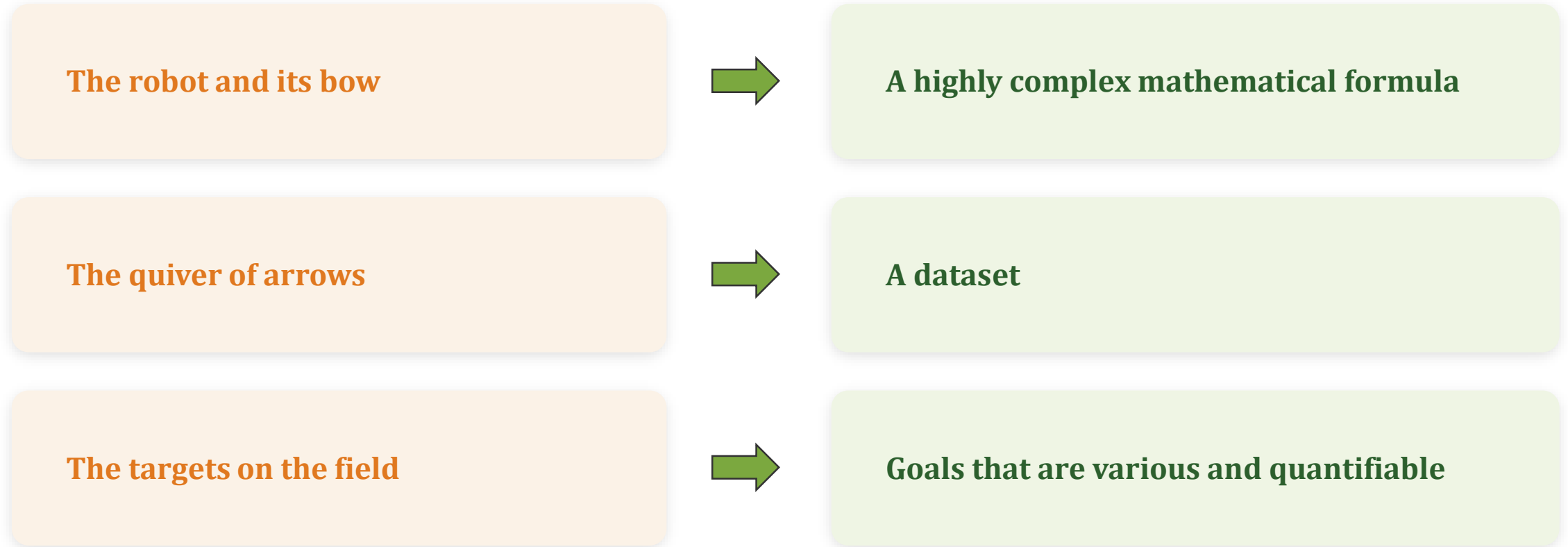
NOTHING ARRIVES

“I fired an arrow and got nothing. Something needs improving before my next shot.”

Positive reinforcement, doing exactly what the name says on the tin.

Now Drop the Archer

The robot was an abstract depiction of what a machine learning model can do. Here is the real mapping.



There are real robots in the world, of course. But the machine learning was never the robot.

The Approaches You Will Actually Meet

The names that come up whenever machine learning is discussed, sorted by where they belong.

Supervised learning

- Support vector machines (SVMs)
- Neural networks
- Deep learning
- Random forest models
- Bayesian networks

Unsupervised learning

- K-means — the most common approach here
- Neural networks, some of which apply here too
- Deep learning

Yes, deep learning is deliberately in both columns. That is not a mistake — it is the reason the next slide exists.

Why Deep Learning Sits in Both Columns

A relatively new computational approach, and the one currently acclaimed as state of the art.

It is genuinely different

Fundamentally unlike the other approaches in how it works, rather than a variation on any of them.

It applies almost everywhere

A broad practical scope across all ML areas, and the reason is simple: the accuracy of its models is extremely high.

Deep learning is still divided into supervised, unsupervised and reinforcement. It solves the same problems as everything else here — just in a conceptually different way.

Coming next: machine learning techniques that lean more towards generative AI than traditional AI.

The Backbone, and What Grew on Top of It

Those three types once operated independently. Hybrid methods have since taken over much of the field.

Supervised

Unsupervised

Reinforcement

New hybrid methods have emerged, and they are increasingly what machine learning actually looks like today.

These fields are wound together far too tightly to keep separate

Business analytics

Data analytics

Data science

Machine learning

Artificial intelligence

It would simply be wrong to tie any one discipline or technique to only one of these boxes.

Natural Language Processing

The subfield of ML that specialises in supervised and unsupervised techniques for human language.

What it does

Understanding, interpreting and generating human language and text, in written and spoken form.

- Translations
- Summarisations
- Text classifications

What it made possible

NLP became the primary discipline behind the tools people now use to talk to computers at all.

- Virtual assistants such as Siri
- Website chatbots
- Online translators

Notice it sits across supervised and unsupervised learning at once. That is the hybrid pattern starting to appear.

Self-Supervised Learning

The NLP technique behind the most significant recent advances of the digital era.

A hybrid approach: supervised and unsupervised learning, combined into one.

Unlabelled data

The model starts with raw data nobody has annotated.

It generates its own labels

The model produces labels straight out of that data.

It learns from them

Those self-made labels become the training signal.

In other words, the model supervises itself through its own data. Which is exactly where the name comes from.

From Self-Supervision to Language Models

This was the cornerstone that produced language models (LMs) and large language models (LLMs).

What LLMs are for

A state-of-the-art approach to text analysis, label generation, and the generation of other kinds of content besides.

Where it led

It paved the way for models like ChatGPT and Gemini, and sharpened their ability to understand and produce language in a natural, human way.

All of it rests on one deceptively small trick

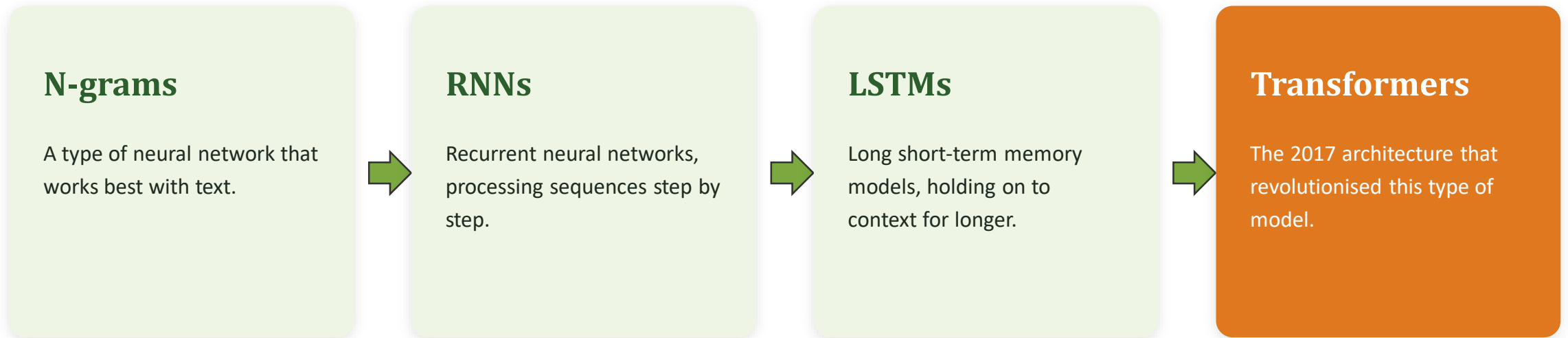
Predicting the missing word in a sentence, using only the context the preceding words provide.

`"The dataset must be cleaned before it can be _____"`

Where LLMs Sit on the Infographic

Large language models are an increasingly popular example of deep learning, built on deep learning architectures.

Historically, deep learning architectures were built like this



So when the infographic puts deep learning under machine learning, this whole lineage is what is sitting inside that one box.

Transformers and the Attention Mechanism

They read text by focusing selectively, rather than handling the whole thing simultaneously.

At each step, the model picks out the parts of the sequence that actually matter for what it is trying to produce.



Darker means the model is weighting that token more heavily right now.

The mechanism

It is called attention, or more precisely self-attention when we are talking about transformers.

What it weighs

The significance of words, parts of words, and other tokens such as subwords or characters, across the input sequence.

Unpacking the Letters in “GPT”

To see how far this technique reaches: the well-known ChatGPT is a transformer.

G

Generative

It creates content, across a genuinely wide range of aspects of life.

P

Pre-trained

It was trained ahead of time, on a vast amount of internet data.

T

Transformer

The architecture from the previous slide, attention mechanism and all.

A generative pre-trained transformer is a large language model — specifically a transformer, and specifically one that has already been trained.

Large Language Models Are Not the Only Option

Generative AI covers several distinct model families, each built for different output.

Diffusion models

Used primarily to generate images and video.

GANs

Generative adversarial networks, aimed at text and images. Distinct because they run two algorithms: one generates the content, the other evaluates the result.

NeRF

Neural radiance fields, which specialise in 3D modelling.

Hybrid models

Any combination of the three above, assembled for the job at hand.

Machine learning is an indispensable component of both generative AI and traditional AI. Supervised, unsupervised and reinforcement learning all turn up in each.

So Where Does AI Meet Data Science?

It is a fair question, given how much ground the last few slides covered.

Imagine you need advice from ChatGPT on how to analyse a specific dataset you have in front of you right now.

Wouldn't you say that is an interaction between AI and data science?

The fields overlap

Business analytics, data analytics, data science, ML and AI cannot be cleanly separated.

The techniques travel

Supervised, unsupervised and reinforcement learning appear across all of them.

The tools meet the work

Every time you use an AI tool on your own data, the two disciplines touch.

Coming next: two real-life examples of machine learning in the wild.

Where ML Shows Up — Fraud Detection

Banks hold enormous datasets of card transactions, and they face fraud every single day.

Why it works: the bank can tell the computer which past transactions were legitimate and which turned out to be fraudulent.

Past transaction	Label
Card used in the home city, midday	Legitimate
Small purchase, familiar merchant	Legitimate
Card used overseas at 3am	Fraudulent
Repeated failed attempts, new device	Fraudulent

That label column is what makes this supervised learning.

Train the model

Supervised learning on that labelled history teaches the model what fraud has looked like before.

Flag in real time

The moment the model sees even the slightest probability of theft, the transaction is flagged and stopped.

Nobody in the sector has reached a perfect solution. The impact of these algorithms has still been groundbreaking.

Where ML Shows Up — Client Retention

Keeping customers is the focus of every business, from a supermarket chain to an online clothing shop.

THE CORNER SHOP

The owner already knows

They recognise their most loyal customers on sight. They offer exclusive discounts to thank them, and those customers keep coming back. No model required.

THE LARGE CHAIN

Nobody can hold it in their head

The bigger a business grows, the harder customer trends are to track. So you pass labelled data to a model and automate exactly what the shop owner was doing by instinct.

What you get out of it

You know which customers are likely to buy, so the discount and the personal touch land where they count — marketing costs down, profit up.

That closes the two biggest rows of the infographic

The main techniques used for working with data and data science, and where each of them is applied.

Traditional data	Big data	Business intelligence	Traditional methods	Machine learning
Collection, pre-processing, balancing, shuffling.	The same backbone, stretched across every data type.	Observations, measures, metrics, KPIs, dashboards.	Regression, clustering, factor analysis, time series.	Supervised, unsupervised, reinforcement, deep learning.

WHAT techniques are involved · WHERE they get applied

Two rows, five columns, and every one of them now has a name you can put to it.